

# Sheth L.U.J. And Sir M.V. College

## Practical No. 14

### 14 Extracting date components using lubridate:: functions (R).

Output:-

```
Console Terminal Background Jobs
R 4.5.2 ~ /
C:\Users\Siddhant\AppData\Local\Temp\KMPKYUSKM\downloaded_packages
> library(lubridate)

Attaching package: 'lubridate'

The following objects are masked from 'package:base':
  date, intersect, setdiff, union

> library(dplyr)

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':
  filter, lag

The following objects are masked from 'package:base':
  intersect, setdiff, setequal, union

> library(readr)
> climate_df <- read_csv("Indian_Climate_Dataset_2024_2025.csv")
Rows: 7310 Columns: 13
— Column specification —
Delimiter: ","
```

```
Console Terminal Background Jobs
R 4.5.2 ~ /
> library(readr)
> climate_df <- read_csv("Indian_Climate_Dataset_2024_2025.csv")
Rows: 7310 Columns: 13
— Column specification —
Delimiter: ","
chr (3): City, State, AQI_Category
dbl (9): Temperature_Max (°C), Temperature_Min (°C), Temperature_Avg (°C), Humidity (%), Rainfall (mm), Wind_Speed (km/h), A...
date (1): Date

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
> str(climate_df)
spec_tbl_ [7,310 x 13] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ Date      : Date [1:7310], format: "2024-01-01" "2024-01-01" "2024-01-01" ...
 $ City      : chr [1:7310] "Mumbai" "Delhi" "Bengaluru" "Chennai" ...
 $ State     : chr [1:7310] "Maharashtra" "Delhi" "Karnataka" "Tamil Nadu" ...
 $ Temperature_Max (°C): num [1:7310] 32.5 25.4 37.2 37.2 27.4 44.4 32.8 40.4 42.3 27.4 ...
 $ Temperature_Min (°C): num [1:7310] 18 10.7 30.8 30.4 17.5 31.6 25.1 33.5 31 15.3 ...
 $ Temperature_Avg (°C): num [1:7310] 25.2 18.1 34 33.8 22.5 38 28.9 37 36.6 21.3 ...
 $ Humidity (%) : num [1:7310] 77.6 84.1 49 34.2 32.2 91.1 83.9 30.4 51.5 79.5 ...
 $ Rainfall (mm) : num [1:7310] 0 0 3.7 9.5 9.1 51.9 0 8.2 0 0 ...
 $ Wind_Speed (km/h): num [1:7310] 3.3 9 6.6 9 9.2 4 5.2 19.7 18.8 14 ...
 $ AQI       : num [1:7310] 259 130 54 176 97 140 345 75 186 73 ...
 $ AQI_Category : chr [1:7310] "Poor" "Moderate" "Satisfactory" "Moderate" ...
 $ Pressure (hPa) : num [1:7310] 1020 1008 1008 993 1008 ...
 $ Cloud_Cover (%) : num [1:7310] 62.1 46 61.3 70 56.9 9.3 12.1 39.1 89.3 7.4 ...
```

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```
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R - R 4.5.2 - ~/
$ Cloud_Cover (%) : num [1:7310] 62.1 46 61.3 70 56.9 9.3 12.1 39.1 89.3 7.4 ...
- attr(*, "spec")=
.. cols(
.. Date = col_date(format = ""),
.. City = col_character(),
.. State = col_character(),
.. `Temperature_Max` (*C) = col_double(),
.. `Temperature_Min` (*C) = col_double(),
.. `Temperature_Avg` (*C) = col_double(),
.. `Humidity` (%) = col_double(),
.. `Rainfall` (mm) = col_double(),
.. `Wind_Speed` (km/h) = col_double(),
.. AQI = col_double(),
.. AQI_Category = col_character(),
.. `Pressure` (hPa) = col_double(),
.. `Cloud_Cover` (%) = col_double()
.. )
- attr(*, "problems")=externalptr>
> processed_data <- climate_df %>%
+ mutate(
+   Actual_Date = ymd(Date),
+   # B. Extract Date Components
+   Year_Num = year(Actual_Date),
+   Month_Num = month(Actual_Date),
+   Month_Name = month(Actual_Date, label = TRUE, abbr = FALSE)
+ )
```

```
Console Terminal Background Jobs
R - R 4.5.2 - ~/
+ Day_Num = day(Actual_Date),
+ Weekday_Num = wday(Actual_Date),
+ Weekday_Name = wday(Actual_Date, label = TRUE, abbr = FALSE),
+ Quarter = quarter(Actual_Date),
+ Day_of_Year = yday(Actual_Date)
+ )
> print("--- Data with Extracted Date Components ---")
[1] "--- Data with Extracted Date Components ---"
> print(head(processed_data))
# A tibble: 6 x 22
  Date City State `Temperature_Max` (*C) `Temperature_Min` (*C) `Temperature_Avg` (*C) `Humidity` (%) `Rainfall` (mm)
  <date> <chr> <chr> <dbl> <dbl> <dbl> <dbl> <dbl>
1 2024-01-01 Mumbai Mahar... 32.5 18 25.2 77.6 0
2 2024-01-01 Delhi Delhi 25.4 10.7 18.1 84.1 0
3 2024-01-01 Bengaluru Karna... 37.2 30.8 34 49 3.7
4 2024-01-01 Chennai Tamil... 37.2 30.4 33.8 34.2 9.5
5 2024-01-01 Kolkata West ... 27.4 17.5 22.5 32.2 9.1
6 2024-01-01 Hyderabad Telan... 44.4 31.6 38 91.1 51.9
# 14 more variables: `Wind_Speed` (km/h) <dbl>, AQI <dbl>, AQI_Category <chr>, `Pressure` (hPa) <dbl>,
# `Cloud_Cover` (%) <dbl>, Actual_Date <date>, Year_Num <dbl>, Month_Num <dbl>, Month_Name <ord>, Day_Num <int>,
# Weekday_Num <dbl>, Weekday_Name <ord>, Quarter <int>, Day_of_Year <dbl>
> current_time <- now()
> print("--- Current Time Extraction ---")
[1] "--- Current Time Extraction ---"
> print(paste("Current Year:", year(current_time)))
[1] "Current Year: 2025"
> print(paste("Current Month:", month(current_time)))
[1] "Current Month: 12"
```

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```
Console Terminal Background Jobs
R 4.5.2 ~ /
Data with Extracted Date Components
> print(head(processed_data))
# A tibble: 6 x 22
  Date      City      State `Temperature_Max` (`C`) `Temperature_Min` (`C`) `Temperature_Avg` (`C`) `Humidity` (%) `Rainfall` (mm)
  <date>    <chr>    <chr>      <dbl>              <dbl>              <dbl>              <dbl>         <dbl>
1 2024-01-01 Mumbai  Mahar...      32.5                18                 25.2              77.6          0
2 2024-01-01 Delhi    Delhi        25.4                10.7              18.1              84.1          0
3 2024-01-01 Bengaluru Karna...      37.2                30.8              34                49            3.7
4 2024-01-01 Chennai  Tamil...      37.2                30.4              33.8              34.2          9.5
5 2024-01-01 Kolkata  West ...      27.4                17.5              22.5              32.2          9.1
6 2024-01-01 Hyderabad Telan...      44.4                31.6              38                91.1         51.9
# i 14 more variables: `wind_Speed` (km/h) <dbl>, `AQI` <dbl>, `AQI_Category` <chr>, `Pressure` (hPa) <dbl>,
# `Cloud_Cover` (%) <dbl>, `Actual_Date` <date>, `Year_Num` <dbl>, `Month_Num` <dbl>, `Month_Name` <ord>, `Day_Num` <int>,
# `Weekday_Num` <dbl>, `Weekday_Name` <ord>, `Quarter` <int>, `Day_of_Year` <dbl>
> current_time <- now()
> print("--- Current Time Extraction ---")
[1] "--- Current Time Extraction ---"
> print(paste("Current Year:", year(current_time)))
[1] "Current Year: 2025"
> print(paste("Current Month:", month(current_time)))
[1] "Current Month: 12"
> print(paste("Current Day:", day(current_time)))
[1] "Current Day: 15"
> print(paste("Current Hour:", hour(current_time)))
[1] "Current Hour: 10"
>
> |
```

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